19

Valuation by Parts

Up to this point, our analysis has focused on single-business companies. But many large companies have multiple business units, each competing in segments with different economic characteristics. For instance, Anglo-Dutch Unilever competes in food and refreshments, personal products, and home-care products. Even so-called pure-play companies, such as Vodafone (mobile telecommunication services) and Amazon (online retail), often have a wide variety of underlying geographical and category segments. This is not just the case for large companies: consider the local bicycle shop that also has an online sales channel.

If the economics of a company's segments are different, you will generate more insights by valuing each segment and adding them up to estimate the value of the entire company. Trying to value the entire company as a single enterprise will not provide much understanding, and your final valuation may be way off the mark. Consider a simple case where a faster-growing segment has lower returns on capital than a slower-growing segment. If both segments maintain their return on invested capital (ROIC), the corporate ROIC would decline as the weights of the different segments change, while the corporate growth rate would steadily increase.

Valuing by parts generates better valuation estimates and deeper insights into where and how the company is generating value. That is why it is standard practice in industry-leading companies and among sophisticated investors. This chapter explains four critical steps for valuing a company by its parts:

1. Understanding the mechanics of and insights from valuing a company by the sum of its parts
2. Building financial statements by business unit—based on incomplete information, if necessary

3. Estimating the weighted average cost of capital (WACC) by business unit
4. Testing the value based on multiples of peers

THE MECHANICS OF VALUING BY PARTS

The most effective way to explore the mechanics of valuing by parts and the insights that can result is to work through a valuation. Exhibit 19.1 details the key financials, value drivers, valuation results, and multiples for each part of ConsumerCo, a hypothetical business. Its parts are four business units, a financial subsidiary, and a nonconsolidated joint venture. To simplify, we kept all future returns and growth rates constant at 2020 levels for each business unit.

All of ConsumerCo's businesses sell products for personal care, but their economics differ widely. The key financials and value drivers in Exhibit 19.1 make this clear. The company's primary business unit, branded consumer products, sells well-known brands in personal care (mainly skin creams, shaving creams, and toothpaste). It generates $2.0 billion in revenues at returns well above its 8.6 percent cost of capital, but mainly in slow-growth, mature markets. Private label, the next-largest business at $1.5 billion in revenues, produces for large discount chains selling products under their own names. This unit is growing faster than the branded-products business, but at far lower returns on capital that barely meet its cost of capital.

The devices business, with $1.25 billion in revenues, sells electronic devices for personal care, such as sun beds, shavers, and toothbrushes, at a very healthy 18.1 percent return on capital, paired with high growth rates. The newly developed organic-products business has $750 million in revenues in premium products made with natural materials. It generates both the highest returns and the highest growth. The $83 million in annual costs for running the corporate center are shown as a separate business unit. Finally, internal revenues, earnings before interest, taxes, and amortization (EBITA), and invested capital are eliminated in the consolidation of ConsumerCo's financials, as the branded-products business buys components from the private-label business unit.

The discounted-cash-flow (DCF) valuation results and multiples in Exhibit 19.1 reflect these differences in size, growth, and ROIC across the businesses. Not surprisingly, the high returns and large scale in branded products lead to the largest valuation ($5,188 million), and the implied multiple of enterprise value (EV) to net operating profit after taxes (NOPAT) is 16.0 times. The private-label business generates almost a third of the company's revenues but contributes only around 10 percent of value ($1,128 million) because of its low returns on capital. Despite its higher growth rate, its

EXHIBIT 19.1 **ConsumerCo: Valuation Summary, January 2020**

	Key financials $ million			Value drivers %													Valuation $ million	Multiples	
	Revenue	EBITA	Invested capital	Revenue growth				Operating margin				ROIC				WACC	DCF value	EV/ NOPAT	EV/ NOPAT
	2020	2020	2020	2018	2019	2020	2020–25	2018	2019	2020	2020–25	2018	2019	2020	2020–25				Peers
Branded products	2,000	500	1,600	1.5	2.5	3.0	3.0	23.0	24.3	25.0	25.0	19.7	19.9	20.3	20.3	8.6	5.188	16.0	15.6
Private label	1,500	143	900	4.6	4.7	5.0	5.0	7.7	8.5	9.5	9.5	8.5	9.3	10.3	10.3	9.1	1,128	12.2	11.7
Devices	1,250	156	563	7.1	7.3	7.5	7.5	11.2	12.0	12.5	12.5	15.8	17.1	18.1	18.1	10.1	1,474	14.5	14.0
Organic products	750	206	488	9.3	9.5	10.0	10.0	27.6	27.3	27.5	27.5	27.6	27.5	27.5	27.5	8.6	3,440	25.7	24.5
Corporate center	–	(83)	806	4.4	5.0	5.4	5.6[1]									9.1	(1,123)	20.9	–
Eliminations	(500)	(2)	(50)														–	–	–
Total operations	5,000	920	4,306	4.5	5.1	5.5	5.7	17.7	18.4	18.4	18.3	12.6	13.2	13.9	14.7		10,107	16.9	
Customer finance																10.5[2]	150[3]	12.1[2]	12.0[2]
Cosmetics joint venture																9.1	609[4]	17.6	17.0
Excess cash																	250		
Gross enterprise value																	11,117		
Debt																	(1,941)[5]		
Equity value																	9,175		

[1] For HQ: growth of HQ costs.
[2] For customer finance: P/E cost of equity.
[3] At equity value, net of debt, in customer finance.
[4] Equity value of minority stake in cosmetics joint venture.
[5] Excluding debt in customer finance: $1,038 million.

EV/NOPAT multiple of 12.2 is lower than that of the branded-products unit. The devices business, with returns well above cost of capital and growth rates exceeding those of private-label products, is valued at $1,474 million and has a multiple of 14.5 times NOPAT. The organic-products business combines high returns with high growth, achieving a value of $3,440 million, second highest after branded products, but at a much higher implied multiple of 25.7 times NOPAT. With headquarters DCF at a negative $1,123 million and no impact on value from eliminations (see later in this chapter), the value of operations for ConsumerCo totals $10,107 million, corresponding to a weighted average multiple of 16.9 times NOPAT.

ConsumerCo's customer-finance subsidiary provides loans for about a quarter of device revenues. It is valued at $150 million (net of $1,038 million of debt), using cash flow to equity discounted at its cost of equity of 10.5 percent (see the next section). The cosmetics joint venture is valued using an enterprise DCF valuation, but only ConsumerCo's 45 percent stake of the equity, valued at $609 million, is included in ConsumerCo's value.

The combined total of ConsumerCo's businesses, including the finance subsidiary, the cosmetics joint venture, and $250 million of excess cash, is $11,117 million. Subtracting $1,941 million of debt (excluding the portion allocated to the finance subsidiary from the company's total debt of $2,980 million) leads to an equity value of $9,175 million.

ConsumerCo's results illustrate why valuation by parts leads to better results. For example, even while all business units are at constant (but different) growth rates and returns on capital, ConsumerCo's overall growth and return continue to change between 2020 and 2025 as the weight of organic products in the portfolio steadily increases. When the economics of business segments differ greatly, it becomes difficult via a purely top-down approach to understand historical patterns and to project future trajectories for a company's returns and growth. If you had conducted a top-down DCF valuation of ConsumerCo as a single business at a constant 2020 ROIC of 13.9 percent and an ongoing growth rate of 5.5 percent, the resulting value would have been 10 percent too low. Also note how large the differences in multiples are across the businesses (from 12.2 to 25.7 times NOPAT) and how the aggregate multiple for the operating enterprise value matches none of the underlying businesses.

The equity value buildup in Exhibit 19.2 illustrates that branded and organic products generate the bulk of the company's value. They also stand out for their market value added—the difference between DCF value and book value of invested capital. For each dollar of invested capital, value creation is the highest in these two business units.

A valuation-by-parts approach offers insights into the sources and drivers of a company's value creation that a purely top-down view cannot reveal. Exhibit 19.3 shows how additional growth or ROIC affects the value of each of the business units. Given the high returns on capital for the organic-products unit, growth through additional investments in that business would create

EXHIBIT 19.2 **ConsumerCo: Equity Value Buildup, January 2020**

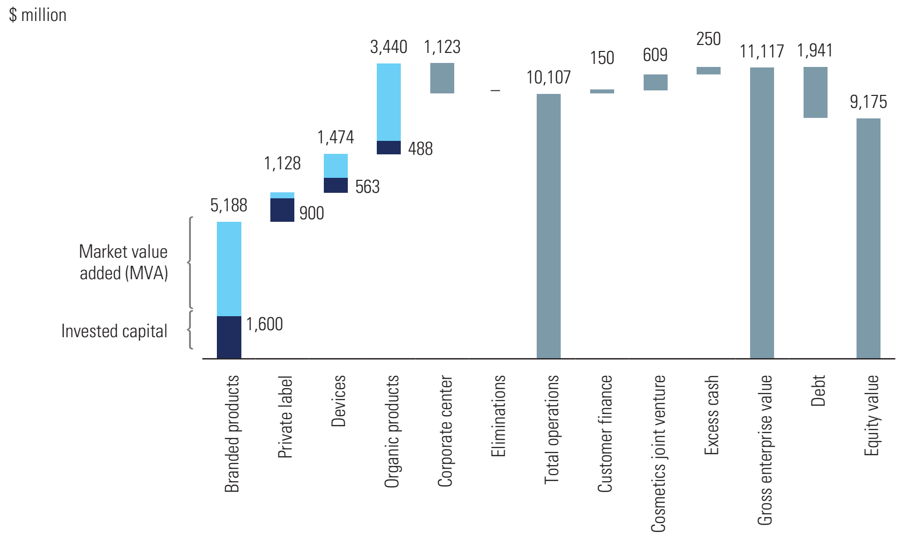
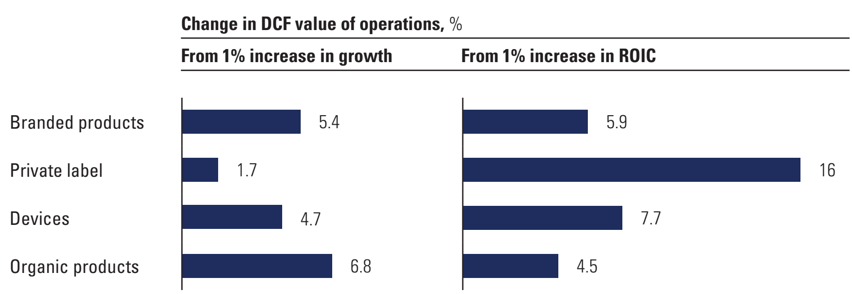


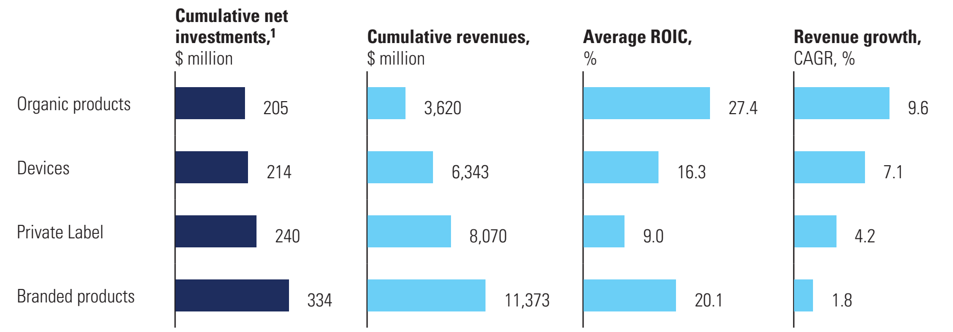
EXHIBIT 19.3 **ConsumerCo: How Changes in ROIC and Growth Affect Value**



more value for the company than investments in other units. In contrast, the private-label unit creates the least amount of value, due to its low returns on capital. Improving returns is the best way to generate more value from this segment. To maximize value creation, ConsumerCo's management should differentiate priorities for growth and return across its segments, rather than set company-wide targets.

Many companies, ConsumerCo among them, struggle with such differentiation. As the investment map in Exhibit 19.4 shows, ConsumerCo's capital expenditures over the five years from 2015 to 2020 have been more in line with the size of each business than with their returns or growth. Investments

EXHIBIT 19.4 **ConsumerCo: Historical Investments, 2015–2020**



[1] Capital expenditures plus investments in net working capital minus depreciation.

were largest in the private-label and branded-products businesses, and lowest in organic products. In the typical annual budgeting process, many companies routinely allocate their capital, research and development (R&D), and marketing budgets to the same activities year after year, regardless of their relative contribution to value creation. The cost is high, since companies that more actively reallocate resources generate, on average, 30 percent higher total shareholder returns (TSR).[1] A valuation by parts can highlight whether a company's capital spending is aligned with its value-creation opportunities.

Sometimes securing the best insights requires even more finely grained valuations than the ConsumerCo example provides. When we analyzed four divisions within a consumer-durable-goods company, we found that all were generating fairly similar returns, between 12 and 18 percent, well above the company's 9 percent cost of capital (see Exhibit 19.5). But at the next level, business units, returns were much more widely distributed. Even in the company's highest-performing division, a business unit was earning returns below its cost of capital. At the level of individual activities within business units, the return distribution was even larger. Differentiating where to invest in growth and where to improve margins at such granular levels can trigger significant improvements in value creation for the company as a whole.[2]

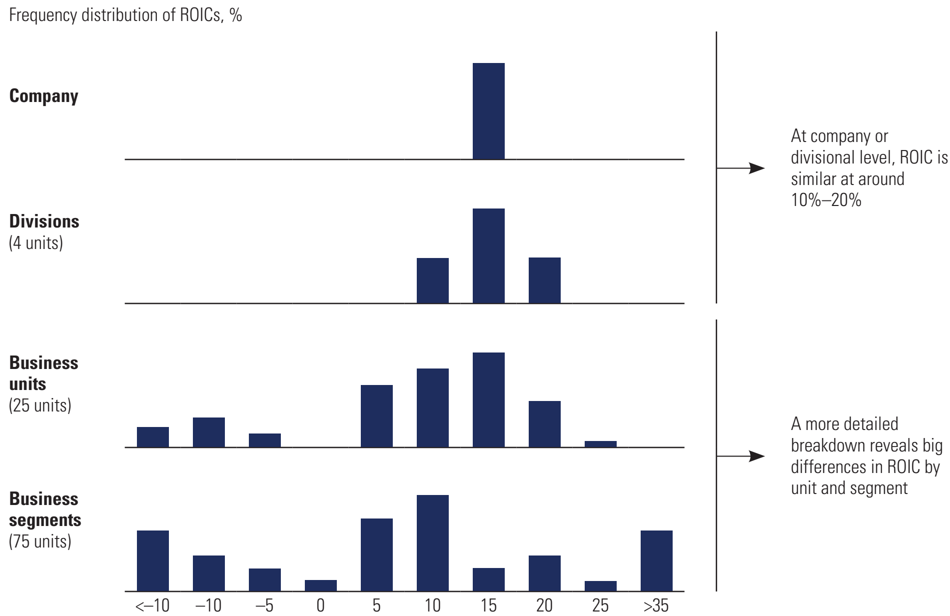
BUILDING BUSINESS UNIT FINANCIAL STATEMENTS

To value a company's individual business units, you need income statements, balance sheets, and cash flow statements. Ideally, these financial statements should approximate what the business units would look like if they

[1] S. Hall, D. Lovallo, and R. Musters, "How to Put Your Money Where Your Strategy Is," *McKinsey Quarterly* (March 2012).

[2] M. Goedhart, S. Smit, and A. Veldhuijzen, "Unearthing the Source of Value Hiding in Your Corporate Portfolio," *McKinsey on Finance* (Fall 2013).

EXHIBIT 19.5 **Breakdown of Return on Invested Capital at Each Level of Analysis**



were stand-alone companies. Creating financial statements for business units requires consideration of several issues:

- Allocating corporate overhead costs
- Dealing with intercompany transactions
- Understanding financial subsidiaries
- Navigating incomplete public information

We will illustrate each of these issues by extending the ConsumerCo example.

Allocating Corporate Overhead Costs

Most multibusiness companies have shared services and corporate overhead, so you need to decide which costs should be allocated to the businesses and which retained at the corporate level. For services that the corporate center provides, such as payroll, human resources, and accounting, allocate the costs by cost drivers. For example, the aggregate cost of human resources services provided by the corporate parent can be allocated by the number of employees in each business unit.

When costs are incurred only because the units are part of a larger company (for example, the CEO's compensation or the corporate art collection), do not allocate the costs. They should be retained as a corporate cost center and valued separately for two reasons. First, allocating corporate costs to business units reduces your ability to compare them with pure-play business unit peers that don't incur such costs (most business units already have their own chief executives, CFOs, and controllers who are comparable to pure-play competitors). Second, keeping the corporate center as a separate unit reveals how much of a drag it creates on the company's value.

For ConsumerCo, the unallocated corporate costs are estimated at $83 million, around 1.7 percent of revenue, with a present value amounting to about 10 percent of enterprise value. The present value of corporate costs is often in the range of 10 to 20 percent of enterprise value for multibusiness companies.

Dealing with Intercompany Transactions

Sometimes business units provide goods and services to one another, incur intragroup payables and receivables, and borrow and lend funds to a group treasury. To arrive at consolidated corporate results, accountants eliminate the internal revenues, costs, and profits, as well as internal assets and liabilities, to prevent double-counting. Only revenues, costs, assets, and liabilities from transactions with external parties remain at the consolidated level. Exhibit 19.6 shows how the 2020 reorganized financials for ConsumerCo's businesses are consolidated with the accounts of the parent company, ConsumerCo Corporation. In this example, ConsumerCo Corporation has no business activities and only holds the equity stakes in the business subsidiaries and most of the group's debt.

Intercompany Sales and Profits ConsumerCo's private-label segment sells partially finished products to the open market but also to the branded-products unit, generating $500 million of internal sales in 2020 (in Exhibit 19.6, see the first line under Eliminations I). If the branded-products unit would process and resell all transferred materials in the same year, $500 million of internal revenues and internal costs could simply be eliminated in the consolidation. Since one unit's revenues are another unit's costs, overall earnings are unaffected.[3]

But, as is often the case for intercompany sales, ConsumerCo's branded-products unit typically does not process and resell all of the private-label deliveries in the same year. Because of the resulting inventory changes of internally

[3] The cumulative value of business units will equal the aggregate value, but the value split depends on the level of transfer pricing between the two units. The higher the transfer price, the more aggregate value is transferred to the private-label business. To value each business unit accurately, record intercompany transfers at the value that would be transacted with third parties. Otherwise, the relative value of the business units will be distorted.

EXHIBIT 19.6 **ConsumerCo: Eliminations and Consolidation, 2020**

$ million

	Subsidiary companies									
	Branded products	Private label	Devices	Organic products	Corporate center	Customer finance	ConsumerCo parent company	Eliminations I	Eliminations II	ConsumerCo consolidated
NOPAT										
Revenues	2,000	1,500	1,250	750	–	–	–	(500)	–	5,000
Operating costs	(1,500)	(1,358)	(1,094)	(544)	(83)	–	–	498	–	(4,080)
EBITA	500	143	156	206	(83)	–	–	(2)		920
Taxes on EBITA	(175)	(50)	(55)	(72)	29	–	–	–	–	(323)
NOPAT	325	93	102	134	(54)	–	–	(2)	–	597
Income from associates and joint ventures	–	–	–	–	–	–	972	–	(942)	30
Interest income	–	–	–	–	–	77	16	–	–	93
Interest expense	–	–	–	–	–	(58)	(118)	–	–	(177)
Taxes on nonoperating items	–	–	–	–	–	(7)	(304)	–	330	19
Net income	325	93	102	134	(54)	12	565	(2)	(612)	563
Invested capital										
Accounts receivable	240	105	25	75	–	–	–	–	–	445
Accounts payable	(216)	(74)	(18)	(5)	–	–	–	–	–	(312)
Inventory	700	375	500	150	–	–	–	(50)	–	1,675
Net PP&E	876	494	55	268	806	–	–	–	–	2,498
Invested capital	1,600	900	563	488	806	–	–	(50)	–	4,306
Excess cash	–	–	–	–	–	–	250	–	–	250
Intercompany receivables	300	–	450	–	–	–	200	–	(950)	–
Loans	–	–	–	–	–	1,154	–	–	–	1,154
Investments in associates and joint ventures	–	–	–	–	–	–	5,097	–	(5,021)	76
Total funds invested	1,900	900	1,013	488	806	1,154	5,547	(50)	(5,971)	5,785
Intercompany payables	–	200	–	–	–	–	750	–	(950)	–
Debt and debt equivalents	–	–	–	–	–	1,038	1,941	–	–	2,980
Adjusted equity	1,900	700	1,013	488	806	115	2,856	(50)	(5,021)	2,806
Total funds invested	1,900	900	1,013	488	806	1,154	5,547	(50)	(5,971)	5,785

supplied materials, one unit's revenues are no longer another unit's costs, and some earnings and inventory now must be eliminated in the consolidation as well. ConsumerCo's consolidated financials eliminate $2 million in earnings and $50 million in inventory (see the Eliminations I column of Exhibit 19.6).[4] As in most situations, the earnings impact is small because it is driven by the change in inventory, not the final inventory. Note that in any case, the eliminations cannot affect ConsumerCo's aggregate free cash flow and enterprise DCF valuation, because consolidation adjustments to inventory always offset the changes in NOPAT.

When you build and forecast the financial statements for the business units, treat each unit as if it were a stand-alone company, using total sales (external plus internal). Otherwise, margins and comparisons over time and with peers will be distorted. Prepare separate projections of the consolidation eliminations, similar to the corporate center. The growth rate of intercompany sales can be estimated from the details of how and why these items arise. It is simplest to assume that the eliminations grow at the same rate as the entire group or as the receiving businesses. Remember, however, that the eliminations are used only to reconcile business unit forecasts to the consolidated-enterprise forecasts. They do not affect the value of the company or the individual business units.

Intercompany Financial Receivables and Payables Multibusiness companies typically manage cash and debt centrally for all business units, which can lead to intercompany receivables from, and payables to, the corporate parent. Sometimes these intercompany accounts are driven by tax considerations. For example, one business unit might lend directly to another unit so that funds don't flow through the parent company, which could trigger additional taxes. Sometimes the accounts have no economic purpose but are simply an artifact of the company's accounting system. Regardless of their purpose, intercompany receivables and payables should not be treated as part of operating working capital but as intercompany equity in the calculation of invested capital.

The Eliminations II column of Exhibit 19.6 shows how this occurs for ConsumerCo. The parent company has $5,097 million of equity investments in its subsidiaries, of which $700 million is in the private-label unit, for example, as reflected in the equity of the subsidiary accounts. This accounting treatment is for internal reports only; since ConsumerCo Corporation owns the private-label business in its entirety, its financial statements are consolidated for external reports, eliminating the $700 million of equity investment. The same holds for the other businesses shown. This leads to the elimination of $5,021

[4] There is no impact on cash taxes or free cash flow from the accounting consolidation. We abstract from any impact of tax consolidation (fiscal grouping) in this example.

million of equity investments in consolidation, leaving only the $76 million stake in the minority-owned cosmetics joint venture as equity investment in the consolidated accounts.

In addition, ConsumerCo Corporation has lent $200 million to the private-label unit, which shows up as an intercompany receivable for the parent company and an intercompany payable for the private-label unit. For the parent company, it represents a nonoperating asset that does not generate operating profits and hence should not be included in its operating working capital. For private label, it represents a financial infusion that is similar to equity. In the consolidated financials, the amounts are eliminated. Similarly, the intercompany receivables for the branded-products and devices businesses are treated as nonoperating assets that are eliminated in the consolidated financials against the $750 million of parent intercompany payables. Failure to handle the intercompany receivables and payables correctly can generate seriously misleading results. In the ConsumerCo example, if the intercompany accounts had been treated as working capital instead of equity, the private-label business's invested capital would have been understated by more than 20 percent, leading to an overstatement of ROIC by roughly the same percentage.

Understanding Financial Subsidiaries

Some firms have financial subsidiaries that provide financing for customers (for example, John Deere Financial and practically all automotive manufacturers). If these subsidiaries are majority owned, they are fully consolidated in the company financial statements. But balance sheets of financial businesses are structured differently from those of industrial or service businesses. The assets tend to be financial rather than physical (largely receivables or loans) and are usually highly leveraged. As detailed in Chapter 38, financial businesses should be valued using cash flow to equity, discounted at the cost of equity. Most companies with significant financial subsidiaries provide a separate balance sheet and income statement for those subsidiaries; the information can be used to analyze and value the financial subsidiaries separately.

Exhibit 19.6 shows that in 2020, ConsumerCo's customer-finance unit has $1,154 million in outstanding customer loans. We estimated the ratio of debt to customer loans required to maintain its current BBB credit rating at 90 percent, so that its funding consists of $1,038 million of debt (0.90 × $1,154 million) and $115 million of equity. The loans generate $77 million in annual interest income. After deducting $58 million of interest expenses on debt and taxes of $7 million, after-tax net income of $12 million remains. The return on equity for the customer-finance unit is 10.8 percent ($12 million of net income divided by $115 million of equity), just above its 10.5 percent cost of equity (see also Exhibit 19.1). These loans, debt, and financial income streams need to be valued separately from ConsumerCo's business operations. Looking ahead, the

EXHIBIT 19.7 **ConsumerCo: Public Information for Business Segments, 2020**

Reported financials, $ million	Branded products	Private label	Devices	Organic products	Corporate center	Intersegment eliminations	Consolidated
Revenues	2,000	1,500	1,250	750	–	(500)	5,000
Operating profit	500	143	156	206	(83)	(2)	920
Depreciation[1]	150	59	57	31	42		338
Capital expenditures	208	107	90	79	42		526
Assets	1,872	882	596	531	830	(50)	4,662
Invested capital: Estimate vs. actual							
Assets/total assets, %	40	19	13	11	18		99
Invested capital estimate, $ million	1,711	806	545	486	759		4,306
Invested capital actual, $ million	1,600	900	563	488	806		4,306
Estimation error, %	6.9	(10.4)	(3.1)	(0.4)	(5.9)		

[1] Included in operating profit.

customer-finance unit's loans are assumed to grow in line with the revenues of the devices business (for which it provides the customer loans). Keeping interest rates and the ratio of debt to customer loans stable at 90 percent, the cash-flow-to-equity DCF value is estimated at $150 million (see Exhibit 19.1).

Be careful not to double-count the debt of the financial subsidiary in the overall valuation of the company. The equity value of the customer-finance subsidiary is already net of its $1,038 million debt, so when we subtracted debt from ConsumerCo's total enterprise value to arrive at the consolidated company's equity value in Exhibit 19.1, we subtracted only the $1,941 million debt associated with business operations.

Navigating Public Information

For our ConsumerCo example, we have the benefit of complete financial statements by business unit. But that will typically not be the case if you are valuing a multibusiness company from the outside in. Exhibit 19.7 shows the disclosure of financial information typical of U.S. Generally Accepted Accounting Principles (GAAP) and International Financial Reporting Standards (IFRS) for a company like ConsumerCo. Companies disclose revenues, operating profit (or something similar, such as EBITA), depreciation, capital expenditures, and assets by segment. You must convert these items to NOPAT and invested capital.

NOPAT To estimate NOPAT, start with reported operating earnings by business unit.[5] Next, allocate operating taxes, any pension adjustment (to

[5] Companies use different names, such as operating profit, underlying profit, or simply earnings before interest and taxes (EBIT), for business unit results.

eliminate the nonoperating effect of pension expense), and operating lease adjustment (eliminating interest expense embedded in rental expense before new accounting standards were introduced in 2019) to each of the business units. (For more information on these adjustments, see Chapter 11.) Use the overall operating tax rate for all business units unless you have information to estimate each unit's tax rate—for example, if units are in different tax jurisdictions. For the ConsumerCo example, this would have resulted in exactly the right NOPAT per business unit, because no pension, lease, or other adjustments are needed on reported EBITA, though this is not typically the case.

After estimating NOPAT, reconcile the sum of all business unit NOPATs to consolidated net income. This step ensures that all adjustments have been properly made.

Invested Capital To estimate invested capital, you can use an incremental approach or a proportional approach, depending on the information available. When possible, use both approaches to triangulate your estimates.

In the *incremental* approach, start with total assets by business unit, and subtract estimates for nonoperating assets and non-interest-bearing operating liabilities. (Note that many companies will hold nonoperating assets at the corporate level, not the unit level. In that case, no adjustment is necessary.) Nonoperating assets include excess cash, investments in nonconsolidated subsidiaries, pension assets, and deferred tax assets. Non-interest-bearing operating liabilities include accounts payable, taxes payable, and accrued expenses. They can be allocated to the business units by either revenue or total assets. As discussed in the earlier section on intercompany payables and receivables, do not treat intercompany loans and debt as an operating liability.

Then allocate the invested capital for the consolidated entity to all of its business units by the amount of total assets minus nonoperating assets and non-interest-bearing liabilities for each business unit. To measure invested capital excluding goodwill,[6] subtract allocated goodwill by business unit. If goodwill is not reported by business unit, you can try to make an estimate from past transactions if these can be aligned with individual business units.

Using the *proportional* approach for ConsumerCo, you could have allocated its total operating invested capital (excluding the customer loans and joint venture, of course) to each of the business units by each unit's proportion of total assets as reported before intersegment eliminations. Note that this would have resulted in some estimation errors, such as allocating $1,711 million invested capital (calculated as $1,872/$4,712 × $4,306 million) to branded products when its true invested capital is $1,600 million.

[6] By goodwill, we mean both goodwill and acquired intangibles.

Once you have estimated invested capital for the business units and corporate center, reconcile these estimates with the total invested capital derived from the consolidated statements.

COST OF CAPITAL

Each business segment should be valued at its own cost of capital, because the systematic risk (beta) of operating cash flows and their ability to support debt—that is, the implied capital structure—will differ by business. To determine an operational business unit's weighted average cost of capital (WACC), you need the unit's target capital structure, its cost of equity (as determined by its levered beta), and its cost of borrowing. For a financial business, you simply need the cost of equity following from its equity beta. (For details on estimating the cost of equity and WACC, see Chapter 15.) The results for ConsumerCo's segments are summarized in Exhibit 19.8.

First, estimate the target capital structure in terms of the debt-to-equity (D/E) ratio for each of ConsumerCo's business units. We recommend using the median capital structure of publicly traded peers, especially if most peers have similar capital structures. Next, determine the levered beta, cost of equity, and WACC. To determine a business unit's beta, first estimate an unlevered median beta for its peer group (be thoughtful about which companies to include, especially outliers). Relever the beta, using the same business unit's

EXHIBIT 19.8 **ConsumerCo: WACC Estimates, January 2020**

Business	Debt/ equity[1]	Cost of debt,[2] %	Beta, unlevered	Beta, levered	Cost of equity,[3] %	WACC,[4] %	DCF value, $ million	Implied debt, $ million
Branded products	0.30	5.5	0.9	1.1	10.1	8.6	5,188	1,197
Private label	0.30	5.5	1.0	1.2	10.7	9.1	1,128	260
Devices	0.25	5.5	1.2	1.5	11.8	10.1	1,474	295
Organic products	0.25	5.5	0.9	1.1	9.9	8.6	3,440	688
Corporate center	0.30					9.4	(1,123)	(259)
Eliminations							–	–
Total operations							10,107	2,181
Customer finance								1,038
Total ConsumerCo: Net debt, implied								3,220
Excess cash, actual								(250)
Debt, actual in operations								1,941
Debt, actual in customer finance								1,038
Total ConsumerCo: Net debt, actual								2,730

[1] At targeted BBB credit rating.
[2] Beta of debt equals 0.2 and risk-free rate of interest equals 4.5%.
[3] Assuming market risk premium of 5.0%.
[4] Tax rate set at 35%.

target capital structure, and estimate its WACC. For the corporate headquarters cash flows, use a weighted average of the business units' costs of capital. Most of ConsumerCo's businesses have similar betas in a range of 1.1 to 1.2, with resulting WACC estimates between 8.6 and 9.1 percent. An exception is the devices business, which is more cyclical at a beta of around 1.5 and a cost of capital of 10.1 percent. For ConsumerCo's customer-finance subsidiary, we directly estimated the equity beta of its peers in retail banking at 1.2, leading to an estimated cost of equity of 10.5 percent.

Finally, using the debt levels based on industry medians, aggregate the business unit debt to see how the total compares with the company's total target debt level.[7] Set the headquarters target D/E at a weighted average of the business units' D/Es, as its negative cash flow is reducing the company's overall debt capacity. If the sum of business unit target debt differs from the consolidated company's actual debt, we typically record the difference as a corporate item, valuing its tax shield separately (or its tax cost when the company is more conservatively financed). Remember that the business units' valuations are based on target, not actual, capital structure.

In ConsumerCo's case, the resulting aggregate target debt level for its business units and finance subsidiary is $3,220 million. That amount is above its total current net debt of $2,730 million, or $2,980 million debt, net of $250 million excess cash (see Exhibit 19.8). If ConsumerCo held on to its current leverage, it would realize a loss in value relative to the value of its parts. To estimate this loss, project the lost tax shields from the company's current below-peer-level leverage into perpetuity at the overall revenue growth rate, and discount these at the unlevered cost of equity.[8]

When you value a company by summing the business unit values, there is no need to estimate a corporate-wide cost of capital or to reconcile the business unit betas with the corporate beta. The individual business unit betas are more relevant than the corporate beta, which is subject to significant estimation

[7] The allocation of debt among business units for legal or internal corporate purposes is generally irrelevant to the economic analysis of the business units. The legal or internal debt is generally driven by tax purposes or is an accident of history (cash-consuming units have lots of debt). These allocations rarely are economically meaningful and should be ignored.

[8] Recall from Chapter 15 that using the cost of debt to discount tax shields significantly overestimates their value. In theory, a company's unlevered cost of equity is a complex average of the unlevered cost of equity of its underlying businesses that changes over time. You can use a simple average of the unlevered costs of equity of the underlying businesses as an approximation, as any associated error has very small impact on value. Assuming a tax rate of 35 percent and an interest rate of 6 percent, the tax shields lost from $490 million in debt below target are $10.3 million for 2020. Assuming future tax shield losses would roughly grow in line with overall revenues, and discounting at ConsumerCo's unlevered cost of equity of around 9.5 percent, the loss in value would amount to around $190 million.

EXHIBIT 19.9 **ConsumerCo: Multiples for Peer Branded-Product Companies, January 2020**

Company	ROIC 2020, %	Growth, 2020–2025, %	EV/EBITA	EV/NOPAT	
Peer 1	31.0	4.7	16.5	22.0	
Peer 2	29.6	4.4	15.8	22.6	*Top-peer average: 21.0*
Peer 3	28.5	4.5	14.3	20.4	
Peer 4	27.1	3.9	12.8	19.1	
Peer 5	22.0	3.0	12.0	16.0	
Peer 6	21.1	2.8	10.5	15.7	
Peer 7	19.7	2.4	11.0	15.7	*Close-peer average: 15.6*
Peer 8	19.0	2.1	10.0	15.4	
Peer 9	18.5	2.2	9.5	15.1	
Peer 10	18.3	4.0	20	26.7	*Outliers*
Peer 11	9.0	3.4	32	45.7	
Overall average				18.0	Excluding outliers
				21.3	Including outliers

error and is likely to change over time as the weights of the underlying businesses in the company portfolio change.[9]

TESTING THE VALUE BASED ON MULTIPLES OF PEERS

Whenever possible, triangulate the discounted cash flow results with valuation multiples, following the recommendations made in Chapter 18. For each of the company's segments, carefully select a group of companies that are comparable not only in terms of sector but also in terms of return on capital and growth. Do not simply take an average or median of the peer group multiples. Instead, always eliminate outliers with multiples that are out of line with their underlying economics, and where possible, estimate a median of close peers with similar returns and growth. Furthermore, we recommend using NOPAT-based instead of EBITA-based multiples, as the latter can be distorted by tax differences across companies.

Exhibit 19.9 shows the EV-to-earnings multiples and underlying ROIC and growth for the competitors of ConsumerCo's branded-products business. Eliminated from the sample are two outliers with valuation multiples that are far above all other peers and not justified by their growth and return on capital. Note how the spread in the EBITA multiples is larger than that of the NOPAT multiples because of different company tax rates. NOPAT multiples are therefore a more reliable basis for the valuation.

[9] The implied cost of capital for ConsumerCo as a whole for each future year is around 8.8 percent. It can be derived by backing it out from the sum of the underlying business units' free cash flows and the sum of the discounted values of these free cash flows.

The overall average NOPAT multiple across the entire peer group is 18.0 times, which would suggest a significantly higher value than the DCF estimate (which has an implied NOPAT multiple of 16.0). But the peers in this group appear to be clustered in two groups with very different underlying returns and growth rates, making the overall average less meaningful. There is a group of leading players with outstanding returns and growth rates that are valued in the stock market at an average of 21.0 times NOPAT. Based on the multiple for this top peer group, ConsumerCo's branded-products business would be valued at $6,883 million, which would be a clear overestimation, given its actual performance and growth (see Exhibit 19.10). At best, it could represent what ConsumerCo's business would be worth if it were able to attain the economics of these leading players in the sector. In contrast, the players in the peer group with returns and growth rates closer to ConsumerCo's business have an average multiple of 15.6 times NOPAT, leading to a value estimate of $5,060 million, which is much closer to the DCF results.

Adopting the same approach of using close-peer multiples to value all of ConsumerCo's other segments, including ConsumerCo finance and the cosmetics joint venture, the estimated equity value is $8,774 million (Exhibit 19.10). Note that by using top-peer multiples for the valuation, ConsumerCo's value would be estimated some 30 percent higher than its DCF value, at $11,956 million. Showing the range of value estimates for close-peer and top-peer multiples helps to triangulate the DCF valuation results. In our experience, close-peer multiples typically lead to valuation results within

EXHIBIT 19.10 **ConsumerCo: Valuation with Multiples, January 2020**

		EV/NOPAT		Multiples-based value				
Business	**NOPAT,** $ million	**Close peers**	**Top peers**	**Close peers,** $ million	**Delta to DCF,** %	**Top peers,** $ million	**Delta to DCF,** %	**DCF value,** $ million
Branded products	325	15.6	21.0	5,060	-2	6,833	32	5,188
Private label	93	11.7	16.0	1,084	-4	1,482	31	1,128
Devices	102	14.0	19.5	1,422	-4	1,980	34	1,474
Organic products	134	24.5	26.5	3,285	-5	3,553	3	3,440
Corporate center	(54)			(1,123)		(1,123)		(1,123)
Eliminations	(2)	–	–	–		–		–
Total operations	597			9,727	-4	12,726	26	10,107
Customer finance	12[1]	12.0[1]	12.0[1]	149	0	149	0	150[2]
Cosmetics joint venture	81	17.0	22.0	589	-3	772	27	609[3]
Excess cash				250		250		250
Gross enterprise value				10,716	-4	13,897	25	11,117
Debt				(1,941)		(1,941)		(1,941)
Equity value				8,774	-4	11,956	30	9,175

[1] For customer finance, P/E and net income are shown.

[2] At equity value, net of debt in customer finance.

[3] At equity value of minority stake in cosmetics joint venture.

10 to 15 percent of the DCF outcomes—in other words, within the normal margin of error for any valuation.

However, many analysts and other practitioners often base their valuations on top-peer multiples. The valuation by parts then easily leads to a conclusion that a company suffers from a so-called conglomerate discount and a recommendation that it should be broken up into parts to unlock the valuation gap versus its peers.

The conclusion is as wrong as the recommendation. The discount simply reflects the fact that compared with its top peers, the company is at a lower valuation level because of lower performance. Splitting up the company does not automatically fix that performance gap (and might not even be needed).

Over the years, practitioners and academics have debated whether a conglomerate or diversification discount exists. In other words, does the market value conglomerates at less than the sum of their parts? Unfortunately, the results are incomplete. There is no consensus about whether diversified firms are valued at a discount relative to a portfolio of pure plays in similar businesses.[10] Some argue that they may even trade at a premium. Among studies that claim a discount, there is no consensus about whether the discount results from the weaker performance of diversified firms relative to more focused firms, or whether the market values diversified firms lower than focused firms.[11] In our experience, however, whenever we have examined a company valued at less than pure-play peers, the company's business units had lower growth and/or returns on capital relative to those peers. In other words, there was a performance discount, not a diversification or conglomerate discount.

SUMMARY

Many large companies have multiple business units, each competing in segments with different economic characteristics. Valuing such companies by their individual parts is standard practice in industry-leading companies and among sophisticated investors. Not only does it generate better valuation results, but it also produces deeper insights into where and how the company is generating value.

To value a company by its parts, you need statements of NOPAT, invested capital, and free cash flow that approximate what the business units would look like if they were stand-alone companies. In preparing such statements,

[10] P. Berger and E. Ofek, "Diversification's Effect on Firm Value," *Journal of Financial Economics* 37 (1995): 39–65; and B. Villalonga, "Diversification Discount or Premium? New Evidence from Business Information Tracking Series," *Journal of Finance* 59, no. 2 (April 2004): 479–506.

[11] A. Schoar, "Effects of Corporate Diversification on Productivity," *Journal of Finance* 57, no. 6 (2002): 2379–2403; and J. Chevalier, "What Do We Know about Cross-Subsidization? Evidence from the Investment Policies of Merging Firms" (working paper, University of Chicago, July 1999).

you likely have to separate out corporate center costs, deal with intercompany transactions, and make a separate equity-cash-flow valuation of any financial subsidiaries. Estimate the weighted average cost of capital for each business unit separately, based on the leverage and the betas of its most relevant peer companies.

To triangulate your DCF estimate, make a multiples-based valuation estimate for each individual unit. Make sure to use a peer group that closely matches the unit's return on capital and growth. In our experience, conclusions that a corporate group suffers from a so-called conglomerate discount are often the result of selecting a peer group with significantly higher returns on capital and growth.

REVIEW QUESTIONS

1. BreakfastCo is a food company with three product divisions: branded cereals, generic cereals, and healthy breakfast options. Given the information in Exhibit 19.11, calculate the present value of the firm's operations. What insights does this analysis give you?

EXHIBIT 19.11 **BreakfastCo: Selected Financial Division Data**

	Branded cereals	Generic cereals	Healthy breakfast
Market data			
Revenue$_{t+1}$, $ million	2,000	1,500	500
NOPAT$_{t+1}$, $ million	600	150	200
Long-term revenue growth, %	4	5	7
Long-term return on invested capital, %	20	11	25
Long-term cost of capital, %	9	10	9

2. You have estimated the net operating profit after tax for each of the divisions in a three-division firm at $40 million, $60 million, and $100 million, respectively, for Divisions A, B, and C. Using these figures, you assume that Divisions A, B, and C contribute 20, 30, and 50 percent, respectively, to overall firm value. Is this an appropriate valuation technique? Why or why not?
3. As CFO, you are trying to allocate investment funds across your three-division firm. You observe the revenues last year as $2.0 billion, $1.5 billion, and $1.5 billion, respectively, for Divisions A, B, and C. Using these figures, you assume that Divisions A, B, and C should have investment budgets of 40, 30, and 30 percent, respectively, of the overall firm's investment budget for the year. What might be problematic about this approach?
4. Using the valuation-by-parts framework, how should one allocate corporate overhead costs to different divisions of a company? Provide an example of an allocation procedure.
5. To finance customer purchases, ATVCo, a manufacturer and seller of all-terrain vehicles, recently started a customer financing unit. ATVCo's income statement and balance sheet for the past year are provided in Exhibit 19.12. Separate ATVCo's income statement and balance sheet into the two segments: manufacturing and the customer financing unit. Assume that equity in the financing subsidiary is the difference between finance receivables and debt related to those receivables. What is the return on invested capital for the manufacturing segment? What is the return on equity (ROE) for the customer financing subsidiary? (For simplicity, assume that the tax rate for both manufacturing and financing is 30 percent. Also, note that the firm had general obligation debt during the year but no longer does at year-end.)

EXHIBIT 19.12 **ATVCo: Income Statement and Balance Sheet**

$ million

Income statement		Balance sheet	
Sales of machinery	1,500	Operating assets	2,200
Revenues of financial products	400	Financial receivables	4,000
Total revenues	1,900	Total assets	6,200
Cost of goods sold	(1,000)		
Interest expense of financial products	(350)	Operating liabilities	400
Total operating costs	(1,350)	General obligation debt	0
		Debt related to customer financing	3,600
Operating profit	550	Stockholders' equity	2,200
Interest expense, general obligation	(80)	Total liabilities and equity	6,200
Income taxes	(141)		
Net income	329		

6. In Question 5, you computed ROE based on an equity calculation equal to the difference between finance receivables and debt related to those receivables. Why might this ROE measurement lead to a result that is too high?
7. What are two best practices for testing the sum-of-the-parts valuation based on multiples of peers? Why are they considered best practices?
8. Comment on the following statement: "Since many firms' valuations by a sum-of-the-parts multiples methodology are greater than the current market valuation of these firms, a breakup of these firms would add value for shareholders."